B. Tech in (CSE) **(SESSION- 2024 onwards)**

Course Code- 105406P Computer Network Lab 0 0 2 1

Perform any ten Experiments

List of Experiments:

1. Write a report that includes a diagram showing the topology, type of connection devices, and speed of the wired and wireless LAN in your organization. Also find out the MAC and IP addresses and the subnet mask of your computer.
2. Install and run a network diagnosis tool such as TCP dump or Wireshark. Start capturing packets on an active interface, open a browser and type the address of your favourite search engine. Wait till the page loads and stop capture. List out the type and number of each type of packets captured.
3. Write a program to create a server that listens to port 53 using stream sockets. Write a simple client program to connect to the server. Send a simple text message "Hello" from the client to the server and the server to the client and close the connection.
4. Write a program to create a chat server that listens to port 54 using stream sockets. Write a simple client program to connect to the server. Send multiple text messages from the client to the server and vice versa. When either party types "Bye", close the connection.
5. Write a program to create a server that listens to port 55 using stream sockets. Write a simple client program to connect to the server. The client should request for a text file and the server should return the file before terminating the connection.
6. Write a program to create a server that listens to port 56 using stream sockets. Write a simple client program to connect to the server. Run multiple clients that request the server for binary files. The server should service each client one after the other before terminating the connection
7. Write a program to create a server that listens to port 57 using stream sockets. Write a simple client program to connect to the server. Run multiple clients that request the server for text files. The server should service all clients concurrently.
8. Write a program to create a server that listens to port 59 using datagram sockets. Write a simple client program that requests the server for a binary file. The server should service multiple clients concurrently and send the requested files in response.
9. Creation of sample network topologies using CISCO packet tracer.
10. Creation of a subnet, assignment of addresses and communication within and outside the subnet.
11. Error Detection using CRC (Python/Java): Implement the Cyclic Redundancy Check (CRC) algorithm for error detection in data transmission.
12. Hamming Code for Error Correction (Python/Java): Implement the Hamming Code algorithm to detect and correct single-bit errors in transmitted data.
13. Sliding Window Protocol (Python/Java): Simulate Stop-and-Wait and Go-Back-N ARQ protocols to demonstrate flow control in a network.
14. Selective Repeat ARQ (Python/Java): Implement the Selective Repeat ARQ protocol for reliable data transmission.
15. CSMA/CD and CSMA/CA Simulation (Python/Java): Simulate Carrier Sense Multiple Access with Collision Detection (CSMA/CD) and Collision Avoidance (CSMA/CA).
16. IPv4 and IPv6 Addressing (Python/Java): Implement a program to validate and classify IP addresses as IPv4 or IPv6.
17. ARP and RARP Implementation (Python/Java): Simulate Address Resolution Protocol (ARP) and Reverse Address Resolution Protocol (RARP).
18. BOOTP and DHCP Simulation (Python/Java): Implement the BOOTP and DHCP mechanisms for dynamic IP address allocation.
19. Routing Algorithms (Python/Java): Implement the Distance Vector Routing (DVR) and Link State Routing (LSR) algorithms.
20. TCP and UDP Simulation (Python/Java): Implement a client-server model to demonstrate TCP

and UDP socket programming.

21. Congestion Control using Leaky Bucket Algorithm (Python/Java): Simulate the Leaky Bucket algorithm for congestion control.
22. Congestion Control using Token Bucket Algorithm (Python/Java): Implement the Token Bucket algorithm to regulate network traffic.
23. FTP and HTTP Simulation (Python/Java): Implement a program to simulate FTP file transfer and send HTTP GET/POST requests.
24. DNS Query Resolution (Python/Java): Write a program to perform DNS resolution and retrieve IP addresses for given domain names.
25. Email Client Simulation (Python/Java): Implement a basic email client that can send and receive messages using SMTP and POP3 protocols.